Year 10 Foundation

REVISION PACK

Progress Check 2

Thursday 5 November 2026

Name: _____

Class: _____

Units in this pack

- Graphs 1
- Number 2
- Shape and Space 1

Graphs 1 · Gradients and straight-line graphs (Lessons 3.1, 3.2, 3.3)

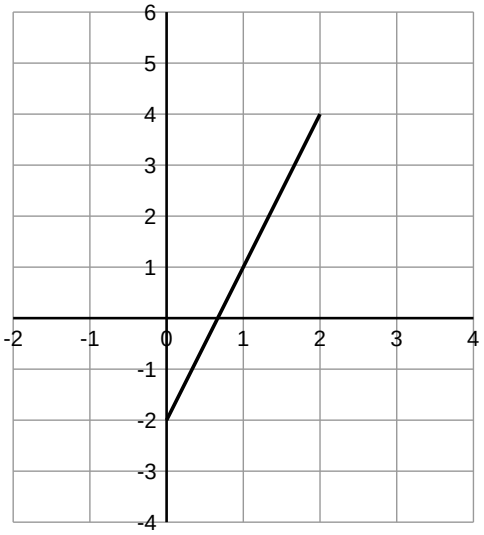
Worked example: gradient from a graph

A straight line is drawn on a grid. It passes through (0, 1) and (2, 7). Find the gradient of the line.

Step 1	Two points	$(0, 1)$ and $(2, 7)$
Step 2	Up	$7 - 1 = 6$
Step 3	Across	$2 - 0 = 2$
Step 4	Up ÷ across	$6 \div 2 = 3$

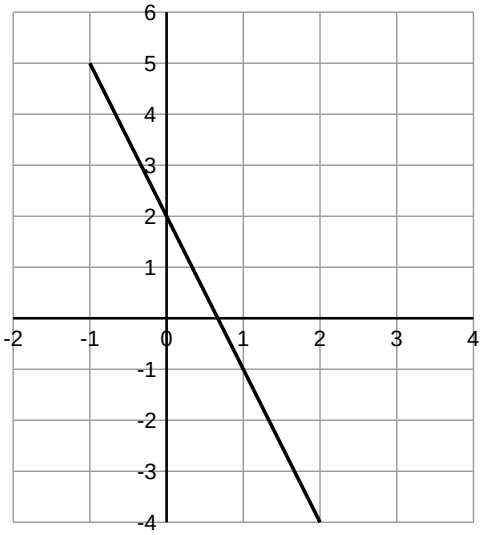
Practice: gradient from a graph

1.1



Find the gradient of the straight line drawn on the grid.

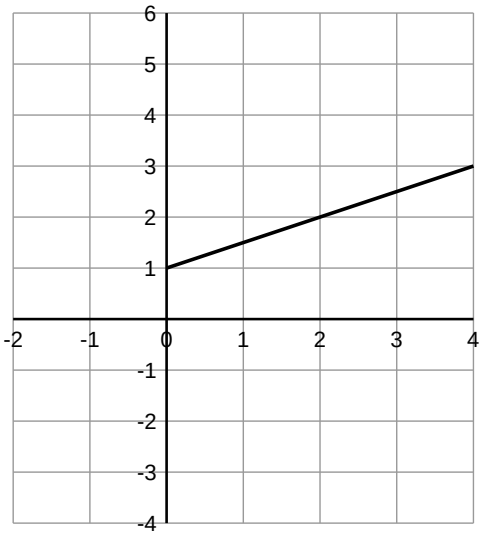
1.2



Find the gradient of the straight line drawn on the grid.

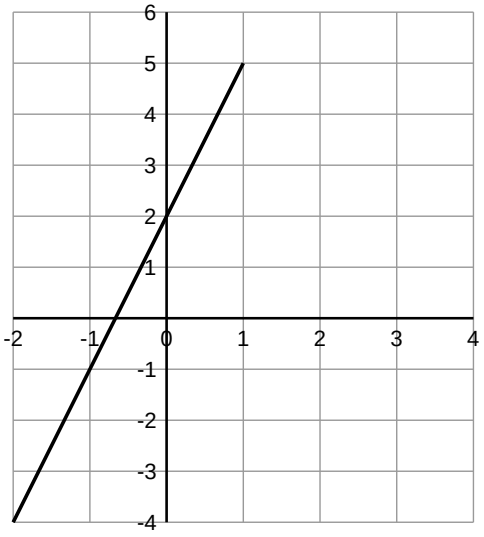
Practice: gradient from a graph (continued)

1.3



Find the gradient of the straight line drawn on the grid.

1.4



Find the gradient of the straight line drawn on the grid.

Graphs 1 · Gradients and straight-line graphs (Lessons 3.1, 3.2, 3.3)

Worked example: gradient from two points

A is the point (2, 1). B is the point (6, 9). Work out the gradient of the line AB.

Step 1	Change in y	$9 - 1 = 8$
Step 2	Change in x	$6 - 2 = 4$
Step 3	Divide	$8 \div 4 = 2$

Practice: gradient from two points

1.5 A is the point (1, 3). B is the point (4, 18). Work out the gradient of the line AB.

1.6 A is the point (2, 10). B is the point (5, 1). Work out the gradient of the line AB.

1.7 A is the point (0, 2). B is the point (8, 6). Work out the gradient of the line AB.

1.8 A is the point (−1, 4). B is the point (3, 16). Work out the gradient of the line AB.

1.9 A is the point (2, 7). B is the point (6, −1). Work out the gradient of the line AB.

1.10 A is the point (1, −2). B is the point (5, 10). Work out the gradient of the line AB.

Graphs 1 · Gradients and straight-line graphs (Lessons 3.1, 3.2, 3.3)

Worked example: drawing a straight-line graph

Here is a table of values for $y = 3x - 1$.

(a) Complete the table of values. (b) Draw the graph of $y = 3x - 1$ on a grid.

(a)	**Step 1**	Put $x = -1$ in	$y = 3 \times (-1) - 1 = -4$
	Step 2	Do each x	$-4,\ -1,\ 2,\ 5,\ 8$
(b)	**Step 1**	Plot the points	$(-1, -4),\ (0, -1),\ (1, 2) \ldots$
	Step 2	Join them	a straight line with a ruler

Practice: drawing a straight-line graph

1.11



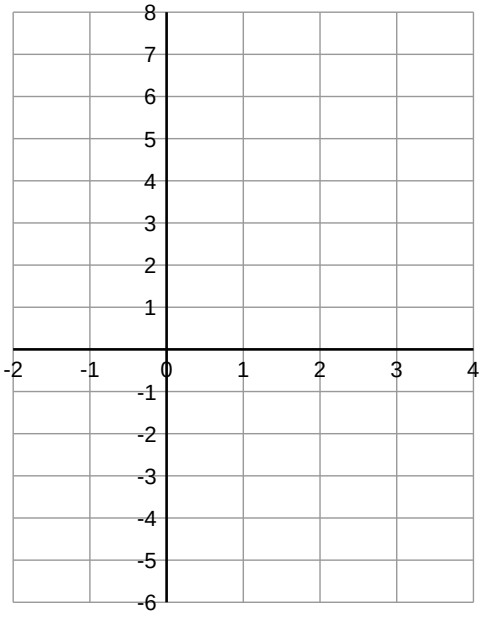
Here is a table of values for $y = 2x + 1$

x	−1	0	1	2	3
y		1		5	

(a) Complete the table of values.

(b) On the grid, draw the graph of $y = 2x + 1$ for values of x from −1 to 3

1.12



Here is a table of values for $y = 3x - 2$

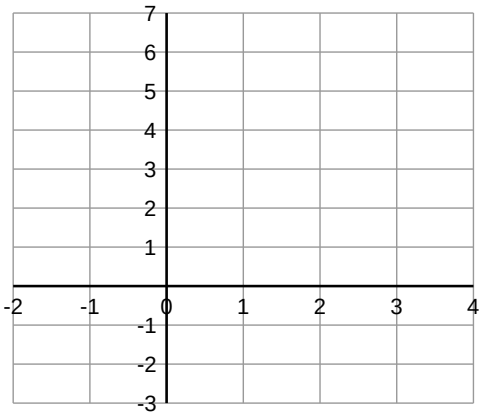
x	−1	0	1	2	3
y		−2		4	

(a) Complete the table of values.

(b) On the grid, draw the graph of $y = 3x - 2$ for values of x from −1 to 3

Practice: drawing a straight-line graph (continued)

1.13



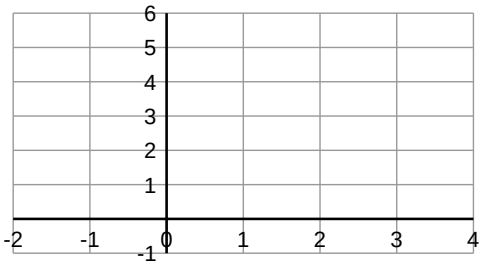
Here is a table of values for $y = -2x + 4$

x	−1	0	1	2	3
y		4		0	

(a) Complete the table of values.

(b) On the grid, draw the graph of $y = -2x + 4$ for values of x from −1 to 3

1.14



Here is a table of values for $y = x + 2$

x	−1	0	1	2	3
y		2		4	

(a) Complete the table of values.

(b) On the grid, draw the graph of $y = x + 2$ for values of x from −1 to 3

Graphs 1 · Gradients and straight-line graphs (Lessons 3.1, 3.2, 3.3)

Worked example: gradient and intercept

A straight line has equation $y = 5 - 2x$

(a) Write down the gradient of the line.

(b) Write down the coordinates of the point where the line crosses the y-axis.

(a)	**Step 1**	$y = mx + c$	$m = -2$
(b)	**Step 1**	c is where it crosses	$c = 5$
	Step 2	Answer	$(0, 5)$

Practice: gradient and intercept

1.15 A straight line has equation $y = 4x - 3$

(a) Write down the gradient of the line.

(b) Write down the coordinates of the point where the line crosses the y-axis.

1.16 A straight line has equation $y = 8 - 5x$

(a) Write down the gradient of the line.

(b) Write down the coordinates of the point where the line crosses the y-axis.

1.17 A straight line has equation $y = 6 - x$

(a) Write down the gradient of the line.

(b) Write down the coordinates of the point where the line crosses the y-axis.

1.18 A straight line has equation $y = 3x + 2$

(a) Write down the gradient of the line.

(b) Write down the coordinates of the point where the line crosses the y-axis.

1.19 A straight line has equation $y = 9 - 4x$

(a) Write down the gradient of the line.

(b) Write down the coordinates of the point where the line crosses the y-axis.

1.20 A straight line has equation $y = 6x - 1$

(a) Write down the gradient of the line.

(b) Write down the coordinates of the point where the line crosses the y-axis.

Number 2 · Percentages (Lesson 4.1)

Worked example: percentage decrease

A TV normally costs KD 360. In a sale, the price is reduced by 20%. Work out the sale price of the TV.

Step 1	20% of 360	$0.2 \times 360 = 72$
Step 2	Subtract	$360 - 72 = 288$
Step 3	Answer	KD 288

Practice: percentage decrease

2.1 A bike normally costs KD 250. In a sale, the price is reduced by 12%. Work out the sale price of the bike.

2.2 A sofa normally costs KD 640. In a sale, the price is reduced by 25%. Work out the sale price of the sofa.

2.3 A pair of shoes normally costs KD 85. In a sale, the price is reduced by 40%. Work out the sale price of the shoes.

2.4 A phone normally costs KD 320. In a sale, the price is reduced by 15%. Work out the sale price of the phone.

2.5 A watch normally costs KD 150. In a sale, the price is reduced by 30%. Work out the sale price of the watch.

2.6 A table normally costs KD 96. In a sale, the price is reduced by 25%. Work out the sale price of the table.

Number 2 · Percentages (Lesson 4.1)

Worked example: percentage loss

The table shows what Huda paid for a phone and what Huda sold it for. Work out Huda's percentage loss.

Bought for	Sold for
KD 300	KD 240

Step 1	Loss	$300 - 240 = 60$
Step 2	Loss ÷ cost × 100	$\frac{60}{300} \times 100 = 20\%$
Step 3	Answer	20%

Practice: percentage loss

2.7 The table shows what Ali paid for a car and what Ali sold it for. Work out Ali's percentage loss.

Bought for	Sold for
KD 5000	KD 4250

2.8 The table shows what Mona paid for a laptop and what Mona sold it for. Work out Mona's percentage loss.

Bought for	Sold for
KD 800	KD 680

2.9 The table shows what Omar paid for a bike and what Omar sold it for. Work out Omar's percentage loss.

Bought for	Sold for
KD 120	KD 102

2.10 The table shows what Fahad paid for a motorbike and what Fahad sold it for. Work out Fahad's percentage loss.

Bought for	Sold for
KD 2400	KD 1800

2.11 The table shows what Laila paid for a TV and what Laila sold it for. Work out Laila's percentage loss.

Bought for	Sold for
KD 650	KD 520

2.12 The table shows what Yousef paid for a car and what Yousef sold it for. Work out Yousef's percentage loss.

Bought for	Sold for
KD 9000	KD 7650

Shape and Space 1 · Angles (Lessons 5.1, 5.2)

Worked example: the exterior angle of a triangle

BCD is a straight line. The two interior opposite angles are 48° and 65°. Work out the size of angle x. Give a reason for your answer.

Step 1	Exterior = two opposite angles	$x = 48 + 65$
Step 2	Answer	$x = 113^\circ$
Step 3	Reason	exterior angle of a triangle

Practice: the exterior angle of a triangle

3.1

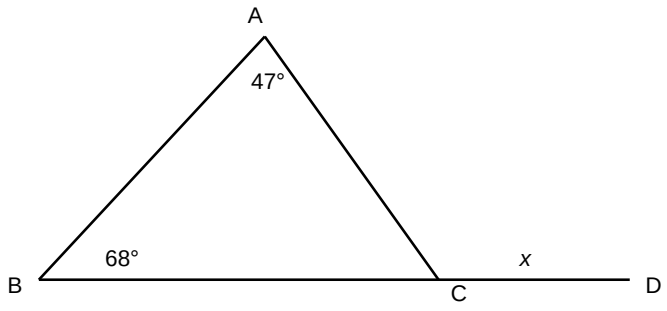


Diagram **NOT** accurately drawn

BCD is a straight line.
Work out the size of angle x.
Give a reason for your answer.

3.2

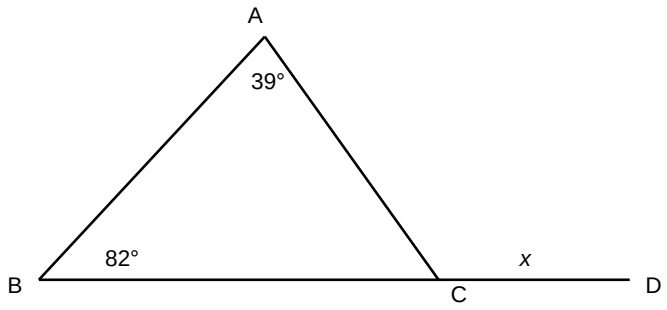


Diagram **NOT** accurately drawn

BCD is a straight line.
Work out the size of angle x.
Give a reason for your answer.

3.3

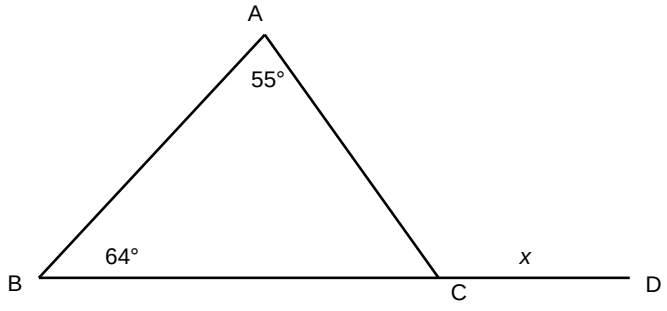


Diagram **NOT** accurately drawn

BCD is a straight line.
Work out the size of angle x.
Give a reason for your answer.

Practice: the exterior angle of a triangle (continued)

3.4

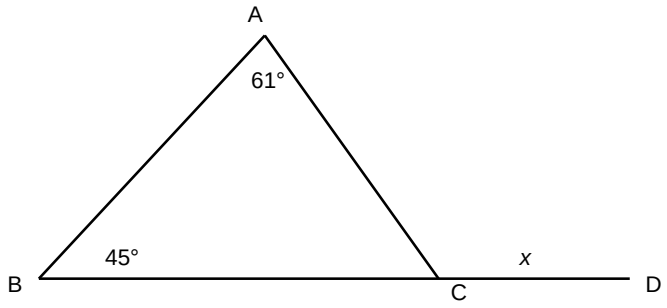


Diagram **NOT** accurately drawn

BCD is a straight line.
Work out the size of angle x.
Give a reason for your answer.

3.5

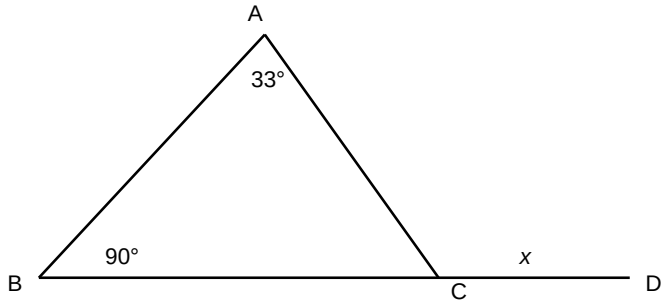


Diagram **NOT** accurately drawn

BCD is a straight line.
Work out the size of angle x.
Give a reason for your answer.

3.6

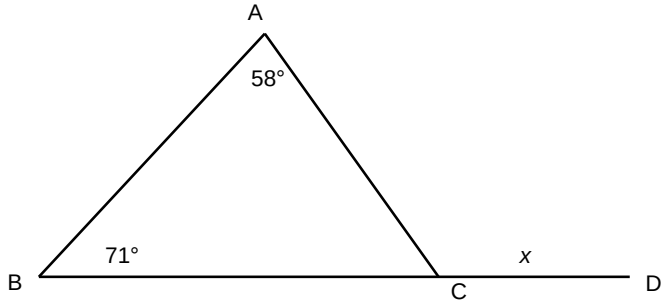


Diagram **NOT** accurately drawn

BCD is a straight line.
Work out the size of angle x.
Give a reason for your answer.

Shape and Space 1 · Angles (Lessons 5.1, 5.2)

Worked example: angles in a quadrilateral

Three angles of a quadrilateral are 95°, 80° and 110°. Work out the size of the fourth angle, x.

Step 1	Angles add to 360°	$95 + 80 + 110 = 285$
Step 2	Subtract	$360 - 285 = 75$
Step 3	Answer	$x = 75^\circ$

Practice: angles in a quadrilateral

3.7

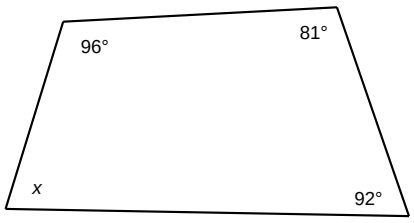


Diagram **NOT** accurately drawn

Work out the size of angle x.

3.8

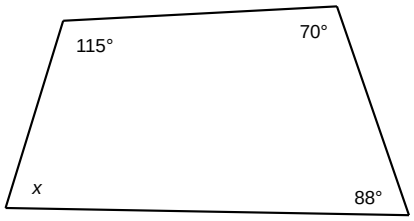


Diagram **NOT** accurately drawn

Work out the size of angle x.

3.9

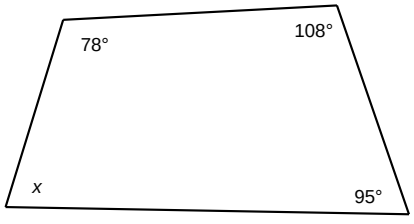


Diagram **NOT** accurately drawn

Work out the size of angle x.

Practice: angles in a quadrilateral (continued)

3.10

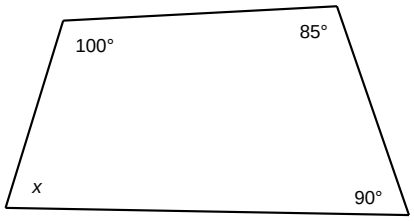


Diagram **NOT** accurately drawn

Work out the size of angle x.

3.11

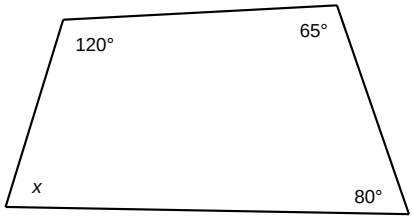


Diagram **NOT** accurately drawn

Work out the size of angle x.

3.12

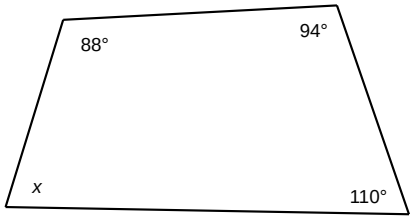


Diagram **NOT** accurately drawn

Work out the size of angle x.

Shape and Space 1 · Pythagoras' theorem, area and circles (Lessons 5.3, 5.4, 5.5)

Worked example: finding the hypotenuse

The two shorter sides of a right-angled triangle are 6 cm and 8 cm. Work out the length of the hypotenuse.

Step 1	Square and add	$x^2 = 6^2 + 8^2 = 100$
Step 2	Square root	$x = \sqrt{100}$
Step 3	Answer	10 cm

Practice: finding the hypotenuse

4.1

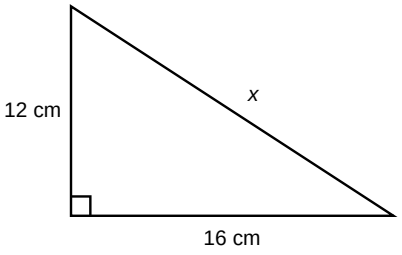


Diagram **NOT** accurately drawn

Work out the length of the hypotenuse, x.

4.2

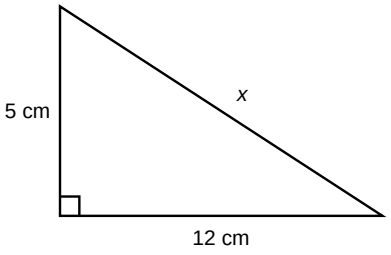


Diagram **NOT** accurately drawn

Work out the length of the hypotenuse, x.

4.3

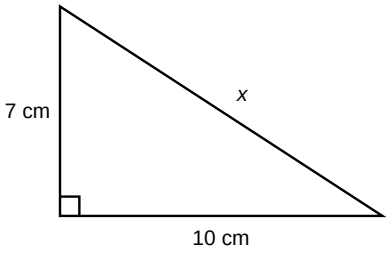


Diagram **NOT** accurately drawn

Work out the length of the hypotenuse, x. Give your answer correct to 3 significant figures.

Practice: finding the hypotenuse (continued)

4.4

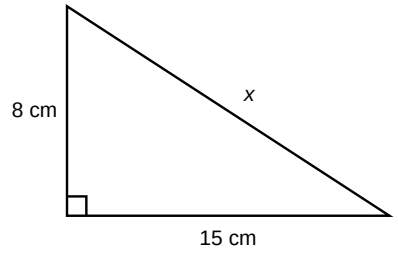


Diagram **NOT** accurately drawn

Work out the length of the hypotenuse, x.

4.5

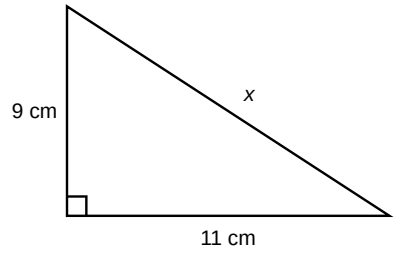


Diagram **NOT** accurately drawn

Work out the length of the hypotenuse, x. Give your answer correct to 3 significant figures.

4.6

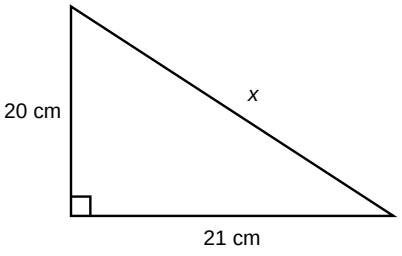


Diagram **NOT** accurately drawn

Work out the length of the hypotenuse, x.

Shape and Space 1 · Pythagoras' theorem, area and circles (Lessons 5.3, 5.4, 5.5)

Worked example: compound shapes

A shape is made from a rectangle 10 cm by 4 cm and a triangle. The triangle has base 10 cm and perpendicular height 3 cm. Work out the total area of the shape.

Step 1	Rectangle	$10 \times 4 = 40$
Step 2	Triangle	$\frac{1}{2} \times 10 \times 3 = 15$
Step 3	Add	$40 + 15 = 55$
Step 4	Answer	$55\ \text{cm}^2$

Practice: compound shapes

4.7

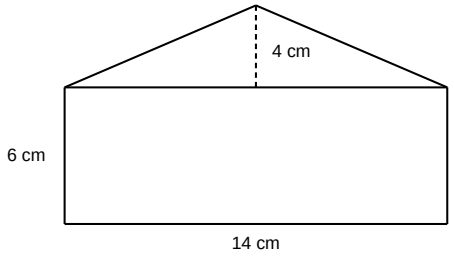


Diagram **NOT** accurately drawn

The shape is made from a rectangle and a triangle.
The triangle has perpendicular height 4 cm.
Work out the total area of the shape.

4.8

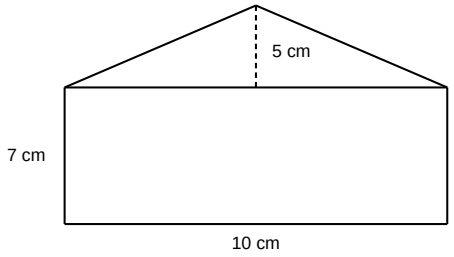


Diagram **NOT** accurately drawn

The shape is made from a rectangle and a triangle.
The triangle has perpendicular height 5 cm.
Work out the total area of the shape.

4.9

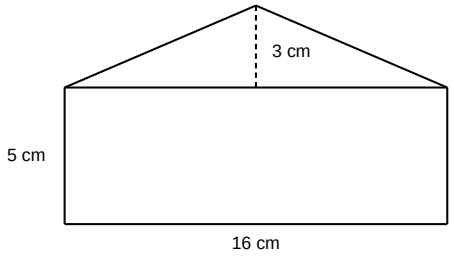


Diagram **NOT** accurately drawn

The shape is made from a rectangle and a triangle.
The triangle has perpendicular height 3 cm.
Work out the total area of the shape.

Practice: compound shapes (continued)

4.10

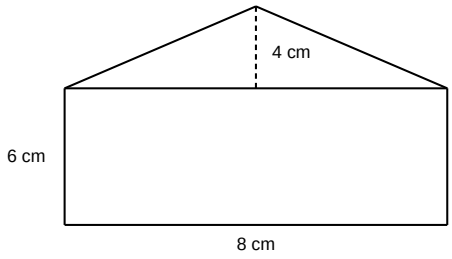


Diagram **NOT** accurately drawn

The shape is made from a rectangle and a triangle.
The triangle has perpendicular height 4 cm.
Work out the total area of the shape.

4.11

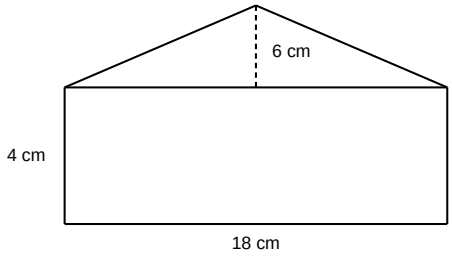


Diagram **NOT** accurately drawn

The shape is made from a rectangle and a triangle.
The triangle has perpendicular height 6 cm.
Work out the total area of the shape.

4.12

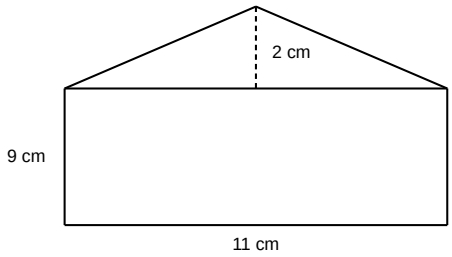


Diagram **NOT** accurately drawn

The shape is made from a rectangle and a triangle.
The triangle has perpendicular height 2 cm.
Work out the total area of the shape.

Worked example: trapezium problems

A lawn is a trapezium with parallel sides 8 m and 12 m and height 5 m. One box of grass seed covers 15 m². How many boxes are needed to cover the lawn?

Step 1	Area	$\frac{1}{2}(8+12)\times 5 = 50\ \text{m}^2$
Step 2	Divide	$50 \div 15 = 3.33\ldots$
Step 3	Round up	4 boxes

Practice: trapezium problems

4.13

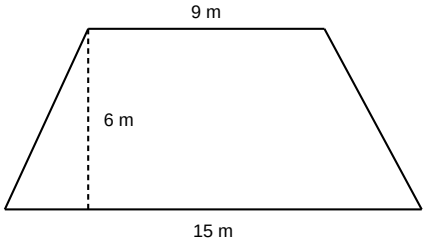


Diagram **NOT** accurately drawn

Ahmad's lawn is a trapezium.
One box of grass seed covers 20 m²
How many boxes must Ahmad buy to cover the lawn?

4.14

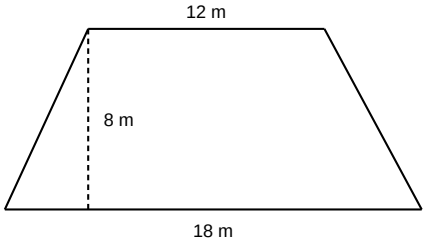


Diagram **NOT** accurately drawn

Noor's lawn is a trapezium.
One box of grass seed covers 25 m²
How many boxes must Noor buy to cover the lawn?

4.15

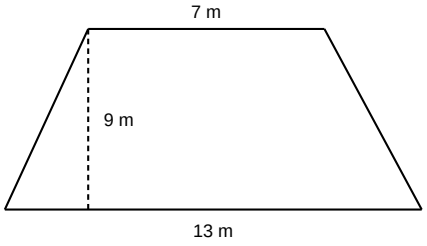


Diagram **NOT** accurately drawn

Faisal's lawn is a trapezium.
One box of grass seed covers 25 m²
How many boxes must Faisal buy to cover the lawn?

Practice: trapezium problems (continued)

4.16

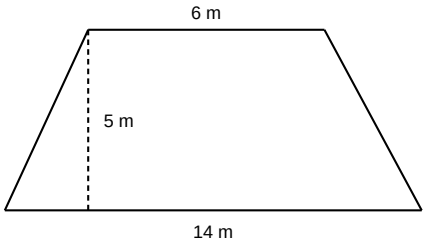


Diagram **NOT** accurately drawn

Sara's lawn is a trapezium.
One box of grass seed covers 15 m²
How many boxes must Sara buy to cover the lawn?

4.17

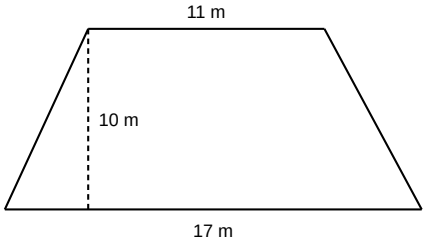


Diagram **NOT** accurately drawn

Yousef's lawn is a trapezium.
One box of grass seed covers 30 m²
How many boxes must Yousef buy to cover the lawn?

4.18

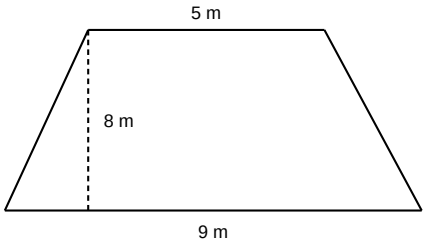


Diagram **NOT** accurately drawn

Maha's lawn is a trapezium.
One box of grass seed covers 10 m²
How many boxes must Maha buy to cover the lawn?

Shape and Space 1 · Pythagoras' theorem, area and circles (Lessons 5.3, 5.4, 5.5)

Worked example: area of a circle

The diameter of a circle is 10 cm. Work out the area of the circle. Give your answer correct to 3 significant figures.

Step 1	Radius	$10 \div 2 = 5$
Step 2	πr^2	$\pi \times 5^2 = 78.5398\ldots$
Step 3	Answer	$78.5\ \text{cm}^2$

Practice: area of a circle

4.19

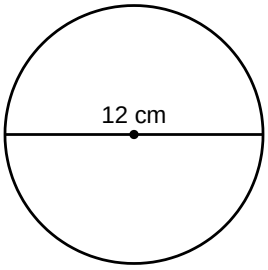


Diagram **NOT** accurately drawn

The diameter of the circle is 12 cm.
Work out the area of the circle.
Give your answer correct to 3 significant figures.

4.20

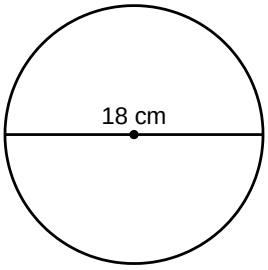


Diagram **NOT** accurately drawn

The diameter of the circle is 18 cm.
Work out the area of the circle.
Give your answer correct to 3 significant figures.

4.21

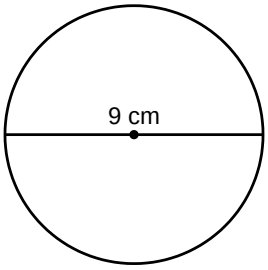


Diagram **NOT** accurately drawn

The diameter of the circle is 9 cm.
Work out the area of the circle.
Give your answer correct to 3 significant figures.

Practice: area of a circle (continued)

4.22

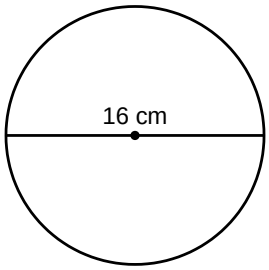


Diagram **NOT** accurately drawn

The diameter of the circle is 16 cm.
Work out the area of the circle.
Give your answer correct to 3 significant figures.

4.23

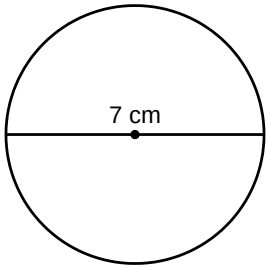


Diagram **NOT** accurately drawn

The diameter of the circle is 7 cm.
Work out the area of the circle.
Give your answer correct to 3 significant figures.

4.24

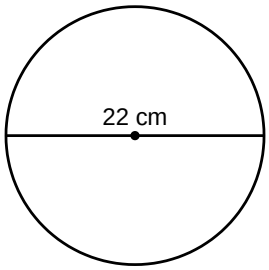


Diagram **NOT** accurately drawn

The diameter of the circle is 22 cm.
Work out the area of the circle.
Give your answer correct to 3 significant figures.